

Class 19: Properties of Determinants

In this section, A is always a square matrix.

1. You get the same value for the $|A|$ no matter which row or column you expand along.

2. If E is any elementary matrix that performs a row operation then

$$|E| = \begin{cases} 1 & \text{if } E \text{ corresponds to row replacement} \\ -1 & \text{if } E \text{ corresponds to a row swap} \\ r & \text{if } E \text{ corresponds to multiplying a row by } r \end{cases}$$

3. $|EA| = |E| |A|$

4. If you row reduce A to its echelon form B using only row swaps and row replacements then $\det A = (-1)^k \det B$ where k is the number of row swaps made.

5. If A is triangular or diagonal then $|A|$ is the product of the diagonal entries.

6. $|AB| = |A| |B|$

7. In general $\det(A+B) \neq \det A + \det B$

8. If c is any scalar then $\det(cA) = c^n \det A$ if A is $n \times n$.

9. $\det(A^T) = \det A$

10. $\det(A^{-1}) = \frac{1}{\det A}$

11. If A is any matrix with linearly dependent columns (or rows) then $|A| = 0$.

12. A is invertible if and only if $\det A$ is nonzero.

Practice

Let A and B be 4×4 matrices with $\det A = -1$ and $\det B = 2$. What is the value for the determinant of each matrix below?

a. $\det (AB) =$

b. $\det (B^5) =$

c. $\det (2A) =$

d. $\det (A^T A) =$

e. $\det (B^{-1}AB) =$

